

ParkScreen — Screening Report

Multimodal decision-aid for Parkinson's disease signs — *not a diagnosis*.

A

OVERALL RISK GRADE

Low likelihood of features consistent with PD

Fused score 0.16

Modality scores

Vocal

weight 87%



Speech channels: phonation 0.46, DDK 0.03.

Facial

weight 13%



Smile classifier 0.00. AU12 peak 4.82, blink 18.7/min.

Vocal channel details

CHANNEL	SCORE	CONFIDENCE	NOTE
Phonation 35% within modality	0.46	1.00	jitter 0.30%, HNR 24.6 dB, F0 SD 1.2 Hz
Articulation (DDK) 65% within modality	0.03	1.00	rate 5.03 syl/s, ISI CV 0.11, amp CV 0.13

Facial channel details

CHANNEL	SCORE	CONFIDENCE	NOTE
Smile task 100% within modality	0.00	1.00	AU12 peak 4.82, blink 18.7/min

Consistency

Channels partially agree

Speech channels align; facial signal is weaker but directionally consistent.

Clinical narrative

Risk Level

Low. The AUC-weighted fused PD probability is **0.157**, indicating that on aggregate the multimodal signals do not resemble the trained PD distribution. As screening evidence, this suggests a low overall likelihood of PD-typical motor-speech and facial signs in this recording, but it does not exclude disease and must be interpreted alongside clinical history.

Phonation

Voice-quality metrics are largely within typical adult ranges: jitter is **0.303%** and shimmer is **3.529%**, both consistent with normal cycle-to-cycle perturbation, and harmonics-to-noise ratio is **24.57 dB**, suggesting a clear phonation with limited noise. However, pitch variability is notably constrained — **f0 SD 1.2 Hz** and **f0 range 7.2 Hz** around a mean of **156.2 Hz** — which can be seen in reduced-prosody presentations, though on a sustained vowel task this may also simply reflect a very steady phonation. The channel model returned a probability of **0.464**, reflecting this mixed picture.

Articulation (DDK)

The PATAKA task shows a **DDK rate of 5.03 Hz**, mildly below the typical adult benchmark of ≥ 6 syllables/sec. Regularity is preserved (**ISI CV 0.112**, **amplitude CV 0.134**), and the amplitude decrement of **-0.0015** is essentially flat, with no meaningful fading across the token sequence. The channel probability of **0.028** reflects an overall articulatory pattern that does not resemble PD-typical dysdiadochokinesia.

Facial Expression

Facial signals are within ranges typical of an expressive face: **AU12 smile amplitude on cue of 4.82** with mean AU12 of **3.00**, **expression variance 0.849**, and **blink rate 18.7/min** (within the ~12–20/min clinical baseline). Head-movement variability is **6.33**, and detection rate was **1.00** with no warnings, so the facial channel is well-supported. The channel probability of **~0.000** shows no hypomimia signature.

Cross-Channel Consistency

Per-channel disagreement — flagged for clinical review. The two speech channels do not agree within threshold: phonation returned an ambiguous **0.464** (driven largely by the low pitch variability on the sustained vowel), while DDK returned a very low **0.028**. The facial channel agrees with the speech-channel mean, so the divergence is confined to phonation vs. DDK. One plausible interpretation is that the sustained-vowel task captured a constricted-prosody signal that the connected-syllable DDK task did not reproduce; alternatively, the phonation score may reflect task-specific steadiness rather than pathology. This split warrants clinician attention despite the low fused probability.

Recommended Next Steps

Given the Low overall risk, no urgent referral is indicated on the basis of this screening alone. However, because of the phonation/DDK disagreement, consider a **repeat recording** with careful attention to sustained-vowel task instructions (natural pitch and loudness variation over a longer duration) to clarify whether the constrained f0 variability persists. If the user has ongoing clinical concerns, **baseline retesting in 6–12 months** would provide a useful longitudinal comparison.

Disclaimers

This report is a screening decision-aid, not a clinical diagnosis. Any risk indication requires evaluation by a qualified movement-disorder specialist.

The classifiers were trained on Parkinson's-disease subjects recorded 2–5 hours post-medication (ON state). Off-state PD may present with more perturbed features than the training distribution, so this system's calibrated probability does not extrapolate to OFF-state inputs.

Screening decision-aid, not a diagnosis. ParkScreen provides supporting evidence for clinical review only. It is not a substitute for evaluation by a qualified healthcare professional and must not be used for self-diagnosis.